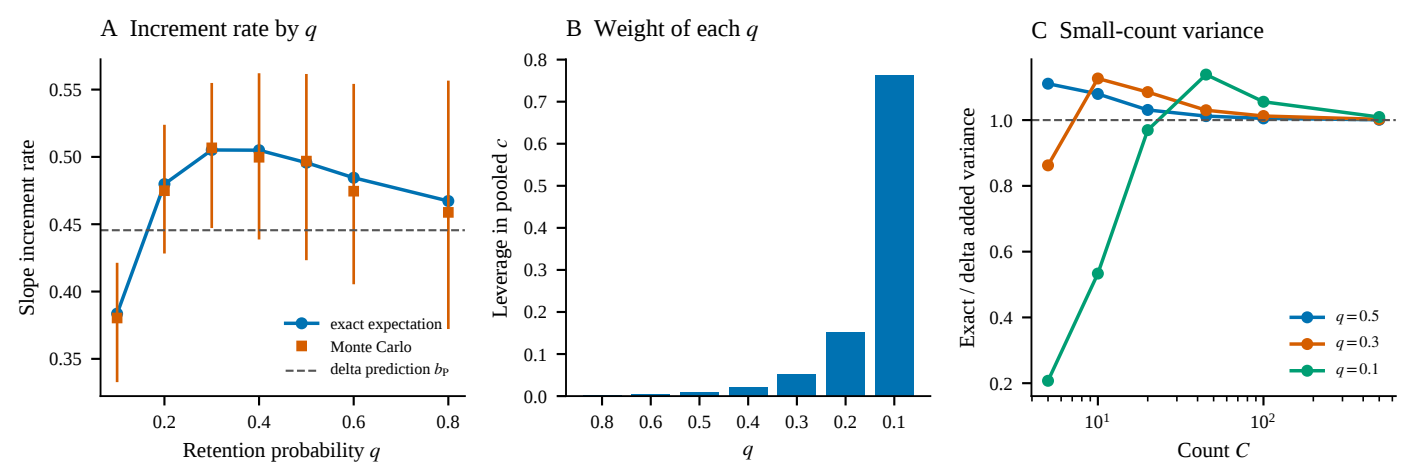
A Increment rate by q
Slope increment rate
0.55
0.50
0.45
0.40
0.35
0.2
0.4
0.6
0.8
Retention probability q
exact expectation
Monte Carlo
delta prediction b_P
B Weight of each q
Leverage in pooled c
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.8
0.6
0.5
0.4
0.3
0.2
0.1
q
C Small-count variance
Exact / delta added variance
1.0
0.8
0.6
0.4
0.2
10^1
10^2
Count C
$q = 0.5$
$q = 0.3$
$q = 0.1$